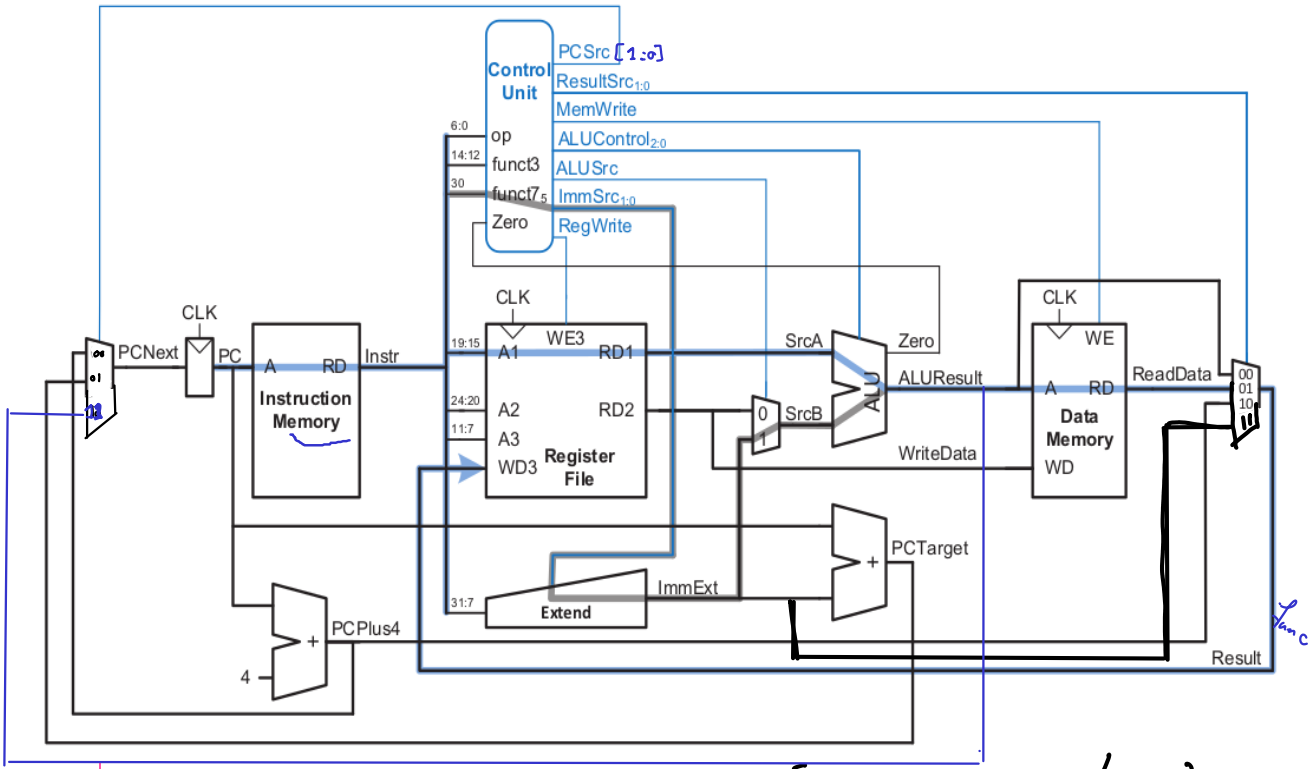
PCSrc[1] =

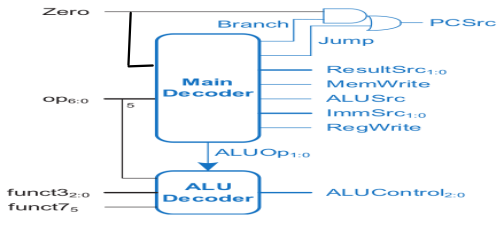


Instruction	Opcode	RegWrite	ImmSrc	ALUSrc	MemWrite	ResultSrc	Branch	ALUOp	Jump	PCSrc[1]
lw	0000011	1	000	1	0	01	0	00	0	0
sw	0100011	0	001	1	1	xx	0	00	0	0
✓ R-type	0110011	1	xxx	0	0	00	0	10	0	0
beq	1100011	0	010	0	0	xx	1	01	0	0
I-type ALU	0010011	1	000	1	0	00	0	10	0	0
jal	1101111	1	011	x	0	10	0	xx	1	0
lui	0110111	1	100	x	0	11	0	xy	0	0
bne	1100011	0	010	0	0	xx	1	01	0	0
jalr	1100111	0	000	1	0	00	0	10	1	1



ALUOp	funct3	{op₅, funct7₅}	ALUControl	Instruction
00	x	x	000 (add)	lw, sw
01	x	x	001 (subtract)	beq
10	000	00, 01, 10	000 (add)	add
	000	11	001 (subtract)	sub
	010	x	101 (set less than)	slt
	110	x	011 (or)	or
	111	x	010 (and)	and

ImmExt = {Instr[31:12], 12'b0}



Address	Instruction	Type	Fields						Machine Language
			$imm_{11:0}$	rs1	f3	rd	op		
0x1000	L7: lw x6, -4(x9)	I	111111111100	01001	010	00110	0000011		FFC4A303
			$imm_{11:5}$	rs2	rs1	f3	$imm_{4:0}$	op	
0x1004	sw x6, 8(x9)	S	0000000	00110	01001	010	01000	0100011	0064A423
			funct7	rs2	rs1	f3	rd	op	
0x1008	or x4, x5, x6	R	0000000	00110	00101	110	00100	0110011	0062E233
			$imm_{12,10:5}$	rs2	rs1	f3	$imm_{4:1,11}$	op	
0x100C	beq x4, x4, L7	B	1111111	00100	00100	000	10101	1100011	FE420AE3

ALUOp	funct3	$\{op_5, funct7_5\}$	ALUControl	Instruction
00	x	x	000 (add)	lw, sw
01	x	x	001 (subtract)	beq
10	000	~~00, 01, 10~~	000 (add)	add
	000	~~11~~	001 (subtract)	sub
	010	x	101 (set less than)	slt
	110	x	011 (or)	or
	111	x	010 (and)	and

~~R type, funct7 == 0~~
~~Jalr~~
~~R type~~ funct7 == 1